

Halloween Decor Assortment Strategy

Case Study: Scenario 2 Recommendation

1. CURRENT ASSORTMENT ASSESSMENT

Items to Keep (In-Store)

Product	Facings	Sim. GM \$	Rationale
12 ft Giant Skeleton	3	\$367,890	Highest profit contributor. Keep at eye level.
7 ft Animated Reaper	1	\$249,560	Highest GM/facing ratio. Small footprint, high return.
9 ft Animated Pirate Ship	2	\$163,423	Hero unit at mid-shelf. Strong price point.
9.5 ft Licensed Sandworm	1	\$88,303	80% sell-through overcomes royalty drag.
5.5 ft LED Skeleton Pony	1	\$94,436	Best margin rate (63.8%) among Giants.

Items to Remove Entirely

Product	Sim. GM \$	Why
LED Fog Machine	\$497	32% sell-through, 5.0% margin. Almost no profit.
12 ft Reaper Inflatable	\$672	50% sell-through, 2.5% margin. 2nd-worst performer.
Halloween Doormat	~\$150	45% sell-through, 17.2% margin. Weakest.

Items to Move to Online-Only

10.5 ft Licensed Character Inflatable - moved to online where fulfillment economics are better. Frees 2 in-store facings for higher-margin items.

2. NEW PRODUCT EVALUATION

Products to Add (In-Store)

Product	Price	Margin %	Why Add
5 ft Skeleton Tombstone	\$149.00	61.2%	High ticket drives \$91+ margin/unit at eye level.
Animated Talking Portrait	\$39.98	82.5%	Highest margin rate. Variety. Waist-level placement.
Solar Pathway Ghosts	\$44.98	64%	Mid-tier decor. No royalty drag. Waist-level.
8 ft Inflatable Black Cat	\$79.00	49.4%	Non-licensed. Waist-level performer.
24 in Light-Up Pumpkin	\$29.98	70%	Impulse buy. Floor-level filler.

Licensed Products & Margin Tradeoff

Keep licensed products that exceed 80% sell-through. The 9.5 ft Licensed Sandworm meets this threshold. Prioritize margin DOLLARS (unit price x margin rate) over margin rate alone.

3. FINAL ASSORTMENT RECOMMENDATION

In-Store Lineup (13 SKUs, 16 Total Facings)

Product	Facings	Position	Category	GM \$/Facing
12 ft Giant Skeleton	3	Eye	Giant	\$122,630
7 ft Animated Reaper	1	Eye	Giant	\$249,560

5 ft Skeleton Tombstone	1	Eye	Decor	\$142,763
9 ft Pirate Ship	2	Mid	Giant	\$81,712
9.5 ft Licensed Sandworm	1	Mid	Inflatable	\$88,303
5.5 ft LED Skeleton Pony	1	Mid	Giant	\$94,436
Animated Talking Portrait	1	Waist	Decor	\$29,748
Solar Pathway Ghosts	1	Waist	Decor	\$22,691
8 ft Inflatable Black Cat	1	Waist	Inflatable	\$24,921
6 ft Scarecrow Inflatable	1	Floor	Inflatable	\$8,147
24 in Metal Tombstone	1	Floor	Decor	\$8,913
Pathway Light Set	1	Floor	Decor	\$9,969
Light-Up Pumpkin Stack	1	Floor	Decor	\$9,378

Category Balance: 6 Giants & Animatronics | 3 Inflatables | 4 Decor & Accessories

Online-Only Items (4 SKUs)

1. Life-Size Animated Butler - \$1,299 | 90% sell-through | 60.4% margin 2. 16 ft Inflatable Haunted Archway - \$699 | 85% sell-through | 48.4% margin 3. Animated Witch Cauldron - \$349 | 85% sell-through | 58.9% margin 4. 10.5 ft Licensed Character (moved from in-store)

Removed Entirely

• LED Fog Machine - \$497 GM, 32% sell-through • 12 ft Reaper Inflatable - \$672 GM, 50% sell-through • Halloween Doormat - 45% sell-through, 17.2% margin

4. PROFIT IMPACT ANALYSIS

Metric	Current Baseline	Scenario 2	Change
Profit per Facing	\$67,961	\$76,259	+12.2%
In-Store Total GM	\$1,087,375	\$1,220,143	+\$132,768
Combined Profit	\$1,484,602	\$1,603,418	+8.0%
Conversion Rate	65.0%	68.0%	+3.0 pts
Avg Items per Customer	1.136	1.239	+9.1%

5. AI REFLECTION

How AI Was Used

1. Monte Carlo Simulation (10,000 customer visits) modeled three shelf scenarios with position multipliers, customer types, and category factors. 2. Shelf Position Optimizer (dynamic programming) tested which 5 items maximize profit per facing at eye level. 3. Data Synthesis extracted historical sell-through, margins, and costs to populate the model.

Key AI Insight That Changed Strategy

The 10.5 ft Licensed Character shows real demand (1,195 simulated units) but 4.2% margin makes it unprofitable in-store. Moving it online frees 2 in-store facings for items earning 14x-60x more profit per facing, while retaining the demand online. This reframing of licensed products by channel directly changed the assortment strategy.

AI Recommendation We Disagreed With

AI initially suggested removing the 9.5 ft Licensed Sandworm due to its 0.85x royalty multiplier. We overrode this: the 80% sell-through rate is an outlier that overcomes the royalty drag. High sell-through = predictable volume. The simulator confirmed: keeping Sandworm (\$88,303 GM/10k visits) beats alternatives. Lesson: Always challenge optimizers when they ignore proven demand in favor of pure margin arithmetic.

What Our Analysis Uncovered That AI Did Not

1. The Margin Rate Paradox - AI ranked by percentage, missing that \$91.35 margin/unit (61.2% on \$149) beats \$32.98/unit (82.5% on \$40). This drove the eye-level repositioning (+12.2% profit/facing). 2. Channel Economics - Licensed items perform differently online (fewer space constraints, different fulfillment). Moving Licensed Character to online preserves revenue while reclaiming in-store space. 3. Customer Journey - Small impulse items must be floor-level for checkout-adjacent value. AI buried them initially.

6. EXECUTIVE SUMMARY FOR VP PRESENTATION

Recommendation

Adopt Scenario 2: Replace the current 16-facing in-store assortment with a 13-SKU lineup optimized for +12.2% profit per facing and +3.0 points conversion rate. Move 1 SKU to online-only; remove 3 SKUs entirely.

Business Impact

In-Store Profit per Facing: +\$8,298 per facing (+12.2%) Total In-Store GM: +\$132,768 (+12.2%) Combined Profit (In + Online): +\$118,816 (+8.0%) Customer Conversion: +3 percentage points Basket Size: +9.1% items per customer

Major Moves

OUT: 3 low-performers (LED Fog, 12ft Reaper, Doormat) combined ~\$1,319 GM IN-STORE: 5 new items simulated ~\$99,147 GM ONLINE: 1 moved SKU (Licensed Character) retains demand, frees 2 high-value in-store facings Net: Shelf reallocated to higher-margin items. Balanced across price points and themes.

Why It Works

1. Proven Heroes Stay: 12ft Skeleton, 7ft Reaper, 9ft Pirate Ship (combined \$780K GM) 2. Smart Removals: Three items simulating <\$700 GM (weak sell-through, low margins) 3. Eye-Level Optimization: Tombstone (high ticket) replaces Talking Portrait, lifting profit +12.2% 4. Channel Discipline: Large items and margin-heavy licensed goods move online 5. Variety: Balanced \$24.98 impulse to \$399+ flagship; spooky, character, outdoor themes